

Chimney Damage Detection

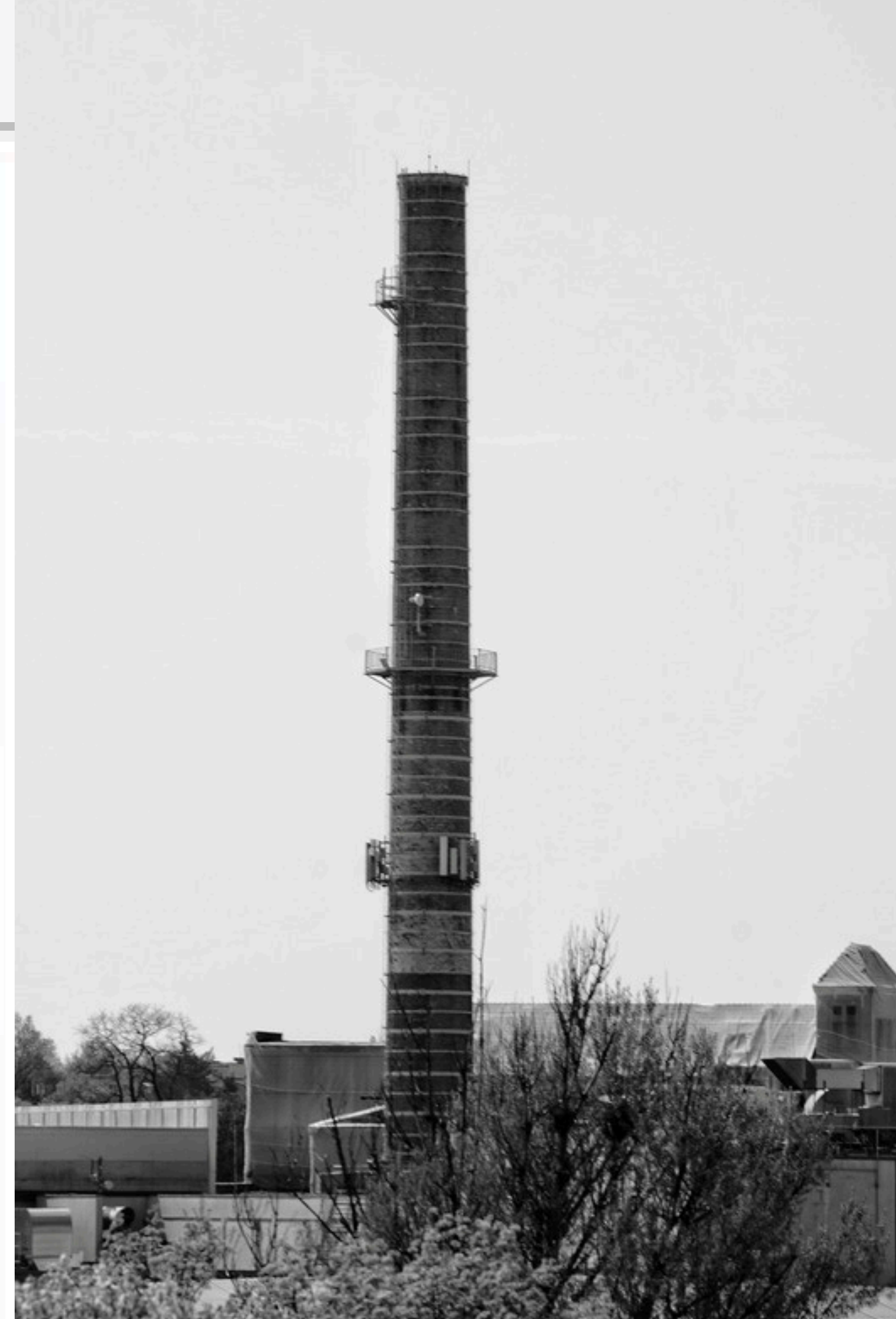
Deep Learning Project using YOLOv8 Segmentation

Presented By

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Presented To

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Problem & Objective

The Problem

Manual chimney inspection is time-consuming, hazardous, and error-prone. Human inspectors struggle to consistently identify subtle structural damage across large structures.

- ❏ Pixel-level segmentation outperforms bounding-box detection by precisely outlining damaged regions — enabling accurate damage quantification.

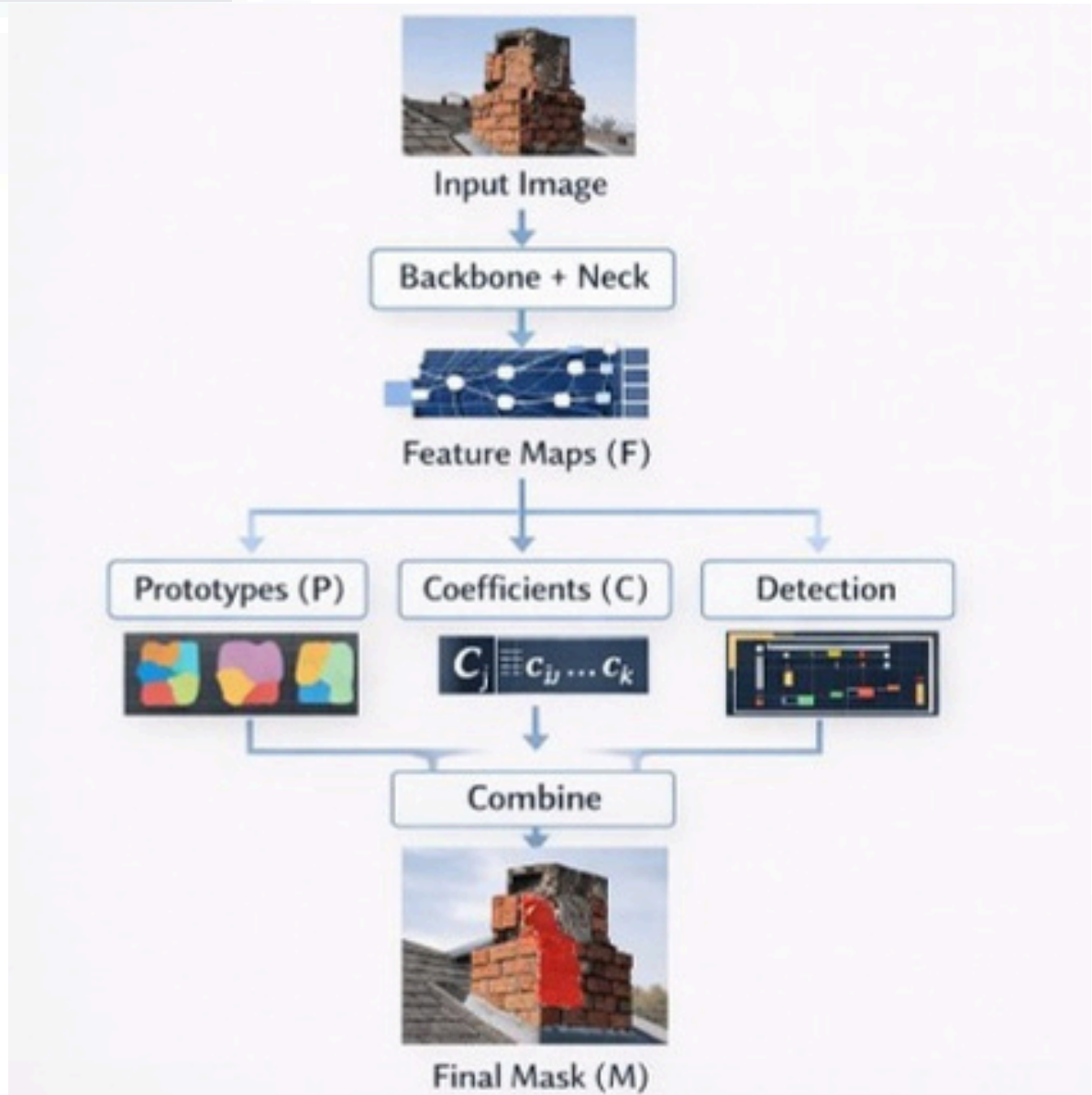
Project Objectives

- Build a YOLOv8 segmentation model
- Detect multiple damage classes
- Evaluate model performance rigorously



YOLOv8 Segmentation Model

ARCHITECTURE OVERVIEW



You Only Look Once (YOLO)

YOLOv8 is the latest generation with improved segmentation heads, anchor-free detection, and superior performance on custom domain tasks.

Mathematical Formulation

- Feature Extraction $F \in \mathbb{R}^{H' \times W' \times C}$
- Prototype Masks $P \in \mathbb{R}^{H' \times W' \times k}$
- Mask Coefficients (per object) $C(j) \in \mathbb{R}^k$
- Final Mask Generation $M_j = \sigma\left(\sum_{i=1}^k C_j \cdot P_i\right)$



Real-Time Speed



High Accuracy



Custom Datasets

Dataset & Annotation

480

Total Images

4

Damage Classes

80%

Train Split

20%

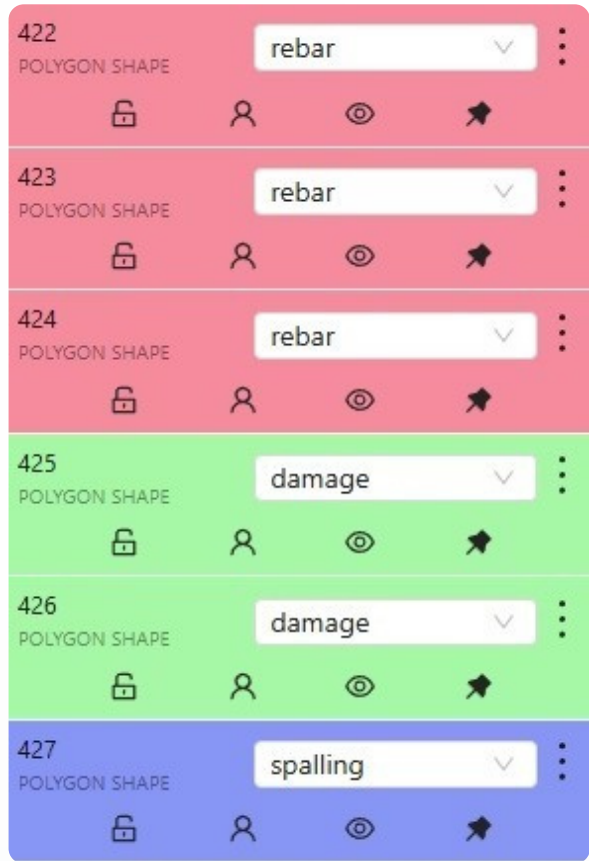
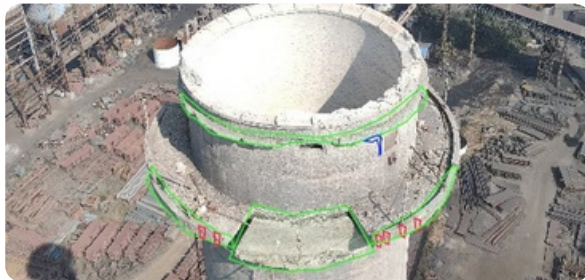
Val Split

 Damage

 Spalling

 Honeycombing

 Exposed Rebars



Annotation Tool: CVAT

Polygon annotations were drawn around each damage region using CVAT (Computer Vision Annotation Tool) for precise, pixel-accurate segmentation masks.

 Google Docs

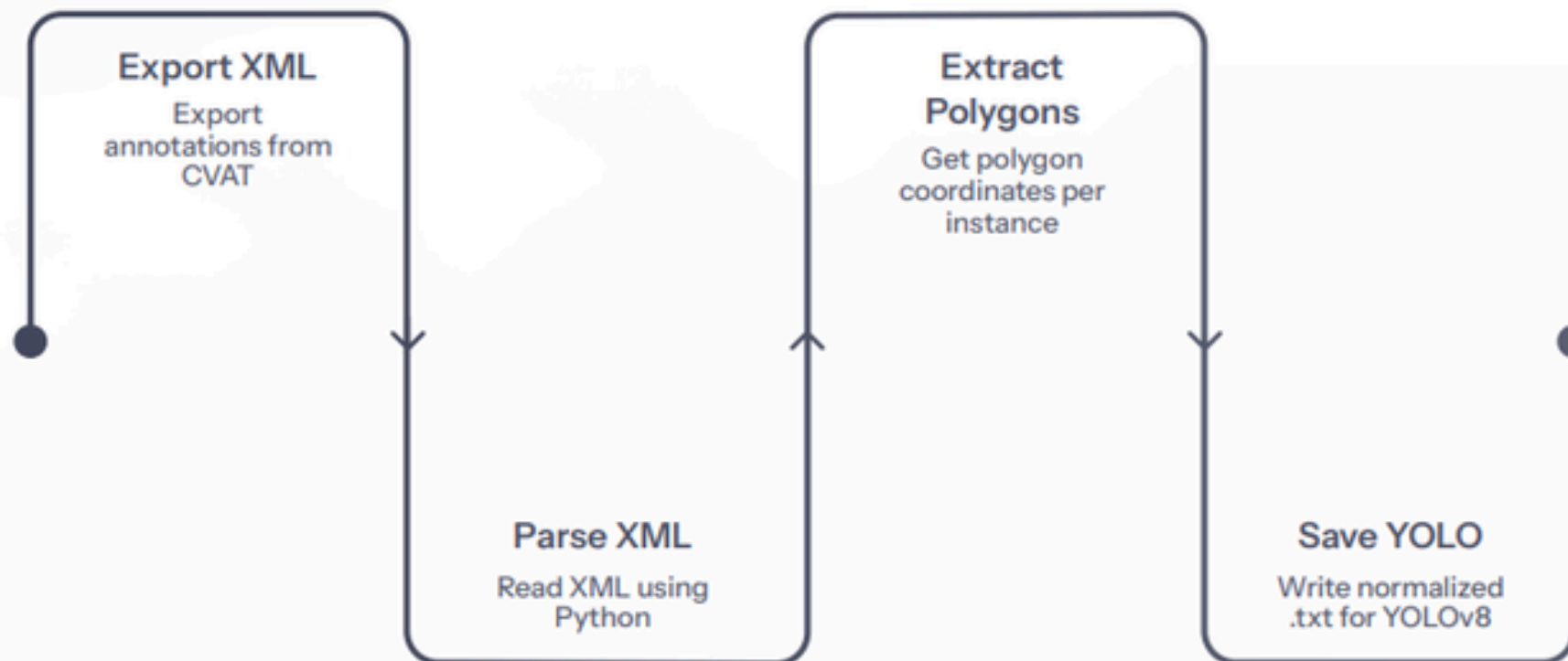
annotations.xml



Data Preparation

CVAT → YOLO FORMAT CONVERSION

Conversion Pipeline



Each label file stores normalized polygon coordinates per line — one row per damage instance — compatible with YOLOv8's segmentation trainer.

Final Dataset Structure

 **images/**

train/ val/

 **labels/**

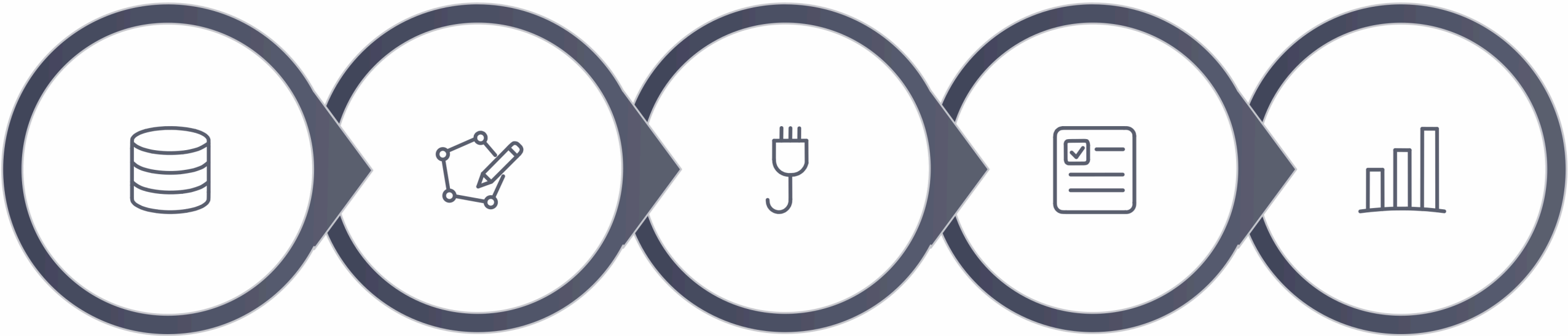
train/ val/

 **data.yaml**

Class names & dataset paths

 The CVAT XML file encodes label names and polygon vertex coordinates per image, which are parsed using Python's `xml.etree` library.

Methodology



Dataset

Annotation

Training

Validation

Testing

Model

YOLOv8n-seg — nano variant optimized for efficient segmentation on limited hardware.

Training Configuration

Image Size: 640	Epochs: 50
Batch Size: 4	Device: CPU

Evaluation Metrics



mAP50

Mean Average Precision at 50% IoU threshold. Measures how accurately the model detects and segments damage regions overall.



Precision

Ratio of correct positive predictions to all predictions made. High precision means fewer false alarms.



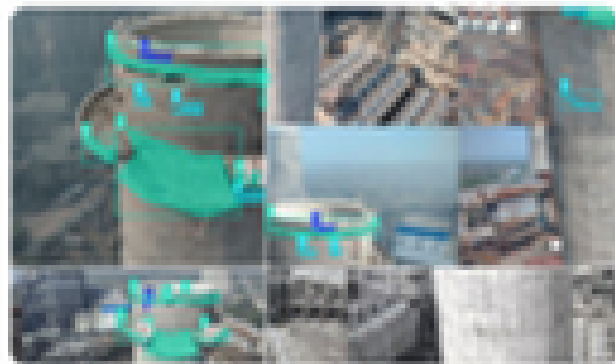
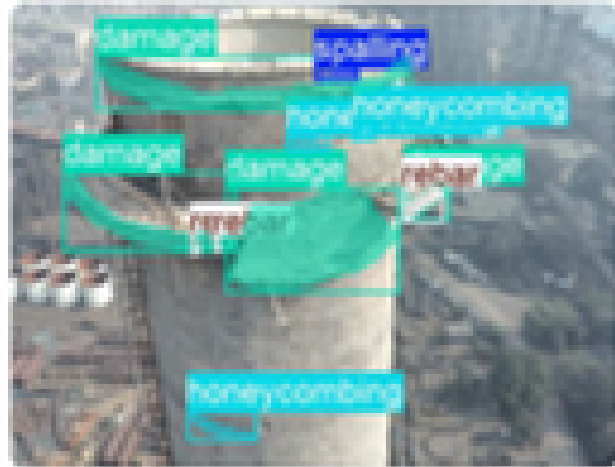
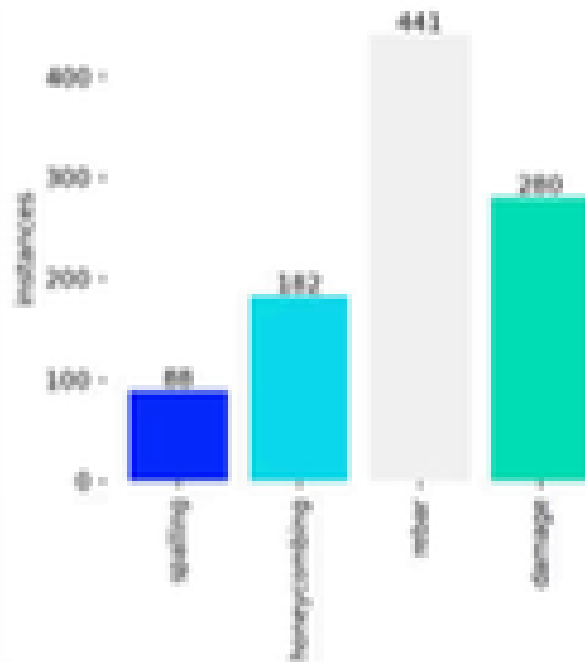
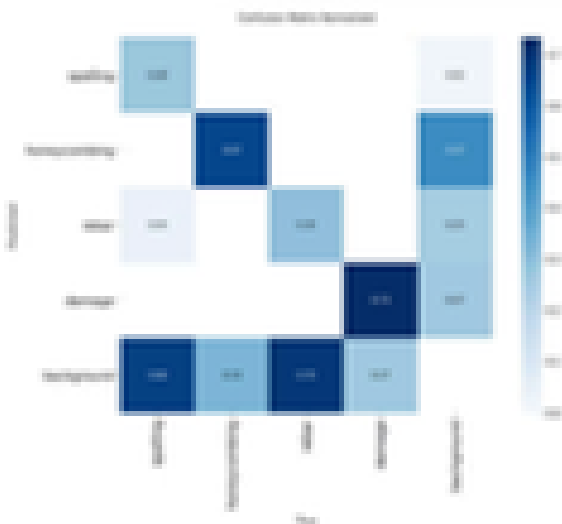
Recall

Ratio of detected objects to all actual objects present. High recall means fewer missed detections.



F1 Score

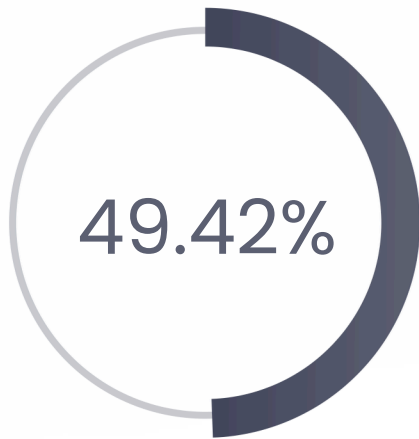
F1 Score gives a balanced measure of how accurate and complete the model's predictions are.



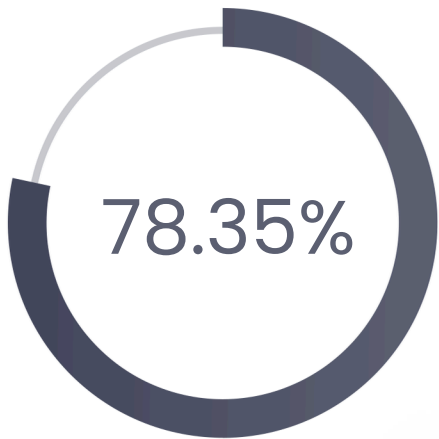
Results

50EPOCHS-YOLOV8N-SEG · CPU TRAINING

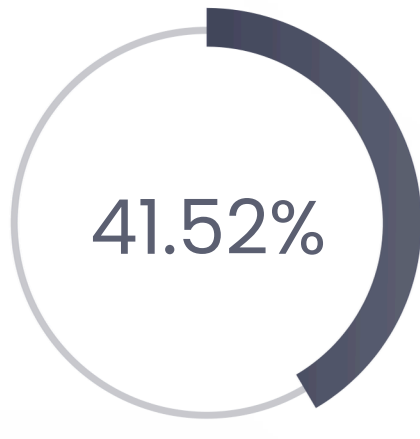
Model Performance



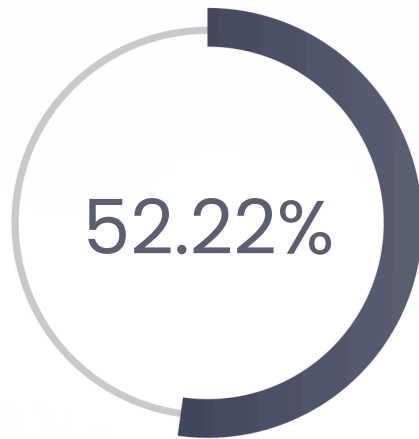
mAP50



Precision



Recall



F1 Score

☐ The model successfully detects all four damage categories — demonstrating strong generalization on a compact 480-image dataset.

